

Setting up a database on AWS

ISTA 322 - Data Engineering

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Networking & Content Delivery


Quantum Technologies


Robotics


Satellite

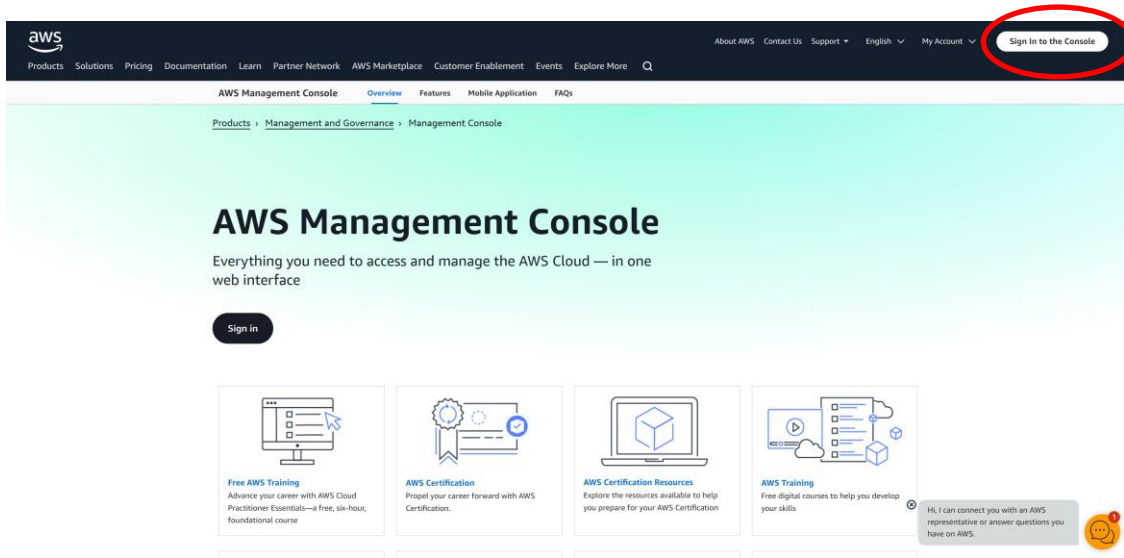

Security, Identity & Compliance


Storage

Go sign up for an AWS account

— — —

<https://aws.amazon.com/console/>



Go sign up for an AWS account

— — —

 **Sign in improvements coming soon**
In our effort to improve the customer experience, Amazon Web Services sign in is getting a new look. [Learn more](#)

×



Sign in

☒ Root user

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ IAM user

User within an account that performs daily tasks. [Learn more](#)

Root user email address

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Signing into AWS

— — —

- <https://aws.amazon.com/console/>
- Sign in as a root user

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Learn more



Sign in

☒ **Root user**

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**

User within an account that performs daily tasks. [Learn more](#)

Root user email address

Next

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Getting to RDB management console

The screenshot shows the AWS Management Console interface. The left-hand navigation pane is visible, listing various AWS services. The 'Database' service is highlighted in the list. The main content area displays the 'Database' console, which lists several database services: Amazon DocumentDB, DynamoDB, ElastiCache, Amazon MemoryDB, Neptune, Amazon QLDB, RDS, and Amazon Timestream. The 'RDS' service is highlighted with a green box and an arrow pointing to it. The right-hand pane shows the 'Applications' section, which is currently empty.

1 - GO TO SERVICES

2 - Click Database

3 - Click RDS

Recently visited

Favorites

All services

Analytics

Application Integration

Blockchain

Business Applications

Cloud Financial Management

Compute

Containers

Customer Enablement

Database

Developer Tools

End User Computing

Front-end Web & Mobile

Game Development

Internet of Things

Machine Learning

Management & Governance

Media Services

Migration & Transfer

Database

Amazon DocumentDB

Fully-managed MongoDB-compatible database service

DynamoDB

Managed NoSQL Database

ElastiCache

In-Memory Cache

Amazon MemoryDB

Fully managed, Redis OSS-compatible, in-memory database service

Neptune

Fast, reliable graph database built for the cloud

Amazon QLDB

Fully managed ledger database

RDS

Managed Relational Database Service

Amazon Timestream

Amazon Timestream is a fast, scalable, and serverless time series database for IoT and operational applications.

Applications (0) Info

Region: US East (N. Virginia)

us-east-1 (Current Region)

Name | Description

Get started

Cost and usage Info

Current month costs

Data unavailable

Forecasted month end costs

Data unavailable

Savings opportunities

Enable Cost Optimization Hub

Creating a database

Click 'Create Database'

Amazon RDS

RDS > Databases

Databases (3)

Group resources     

Filter by databases

	DB identifier ▲	Status ▼	Role ▼	Engine ▼	Region & AZ ▼	Size ▼	Recommendations ▼	CPU ▼	Current activity ▼	Maintenance ▼
	moviesdb	Available	Instance	PostgreSQL	us-east-1f	db.t3.micro	 3 Informational	 3.35%	 0 Connections	none
	nb6coviddata	Available	Instance	PostgreSQL	us-east-1a	db.t3.micro		 3.52%	 0 Connections	none
	ticketsdb	Available	Instance	PostgreSQL	us-east-1c	db.t3.micro	 3 Informational	 3.49%	 2 Connections	none

Creating a database - selecting options

Click 'Standard Create'

Click 'PostgreSQL'

Use default 'PostgreSQL'

Create database

Choose a database creation method [Info](#)

☒ Standard create
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☐ MySQL


☐ MariaDB


☒ PostgreSQL


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Engine version [Info](#)
View the engine versions that support the following database features.

▼ Hide filters

☒ Show versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

Engine Version
PostgreSQL 16.3-R2 ▼

☐ Enable RDS Extended Support [Info](#)

Creating a database - selecting options

- Now we need to give it a name, password, username, etc
- Also select how much compute resources to use
- **Really really important to follow directions**

Creating a database - selecting options

Click 'Free tier'

Create instance id.

NOTE: This is just the name in the console, not for accessing

Create username and password.

NOTE: You'll need these credentials to access db from python.

Templates
Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

☒ **Free tier**
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

Deployment options [Info](#)
The deployment options below are limited to those supported by the engine you selected above.

☒ Multi-AZ DB Cluster
Creates a DB cluster with a primary DB instance and two readable standby DB instances, with each DB instance in a different Availability Zone (AZ). Provides high availability, data redundancy and increases capacity to serve read workloads.

☐ Multi-AZ DB instance (not supported for Multi-AZ DB cluster snapshot)
Creates a primary DB instance and a standby DB instance in a different AZ. Provides high availability and data redundancy, but the standby DB instance doesn't support connections for read workloads.

☐ Single DB instance (not supported for Multi-AZ DB cluster snapshot)
Creates a single DB instance with no standby DB instances.

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Creating a database - selecting options

CRITICAL - Change
'Public access' to Yes.

CRITICAL - Leave 'VPC
security group' as
'Choose existing.'
We'll modify that later.

Expand 'Additional
configuration'

Connectivity [Info](#)

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-09e48b97a597b7b97)
6 Subnets, 6 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default-vpc-09e48b97a597b7b97
6 Subnets, 6 Availability Zones

Public access [Info](#)
☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

default X

Availability Zone [Info](#)

No preference

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
☐ **Create an RDS Proxy** [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Creating a database - selecting options

You can see the port is set to 5432 - the standard for Postgres

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Existing VPC security groups

Choose one or more options

default X

Availability Zone [Info](#)

No preference

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☐ Create an RDS Proxy [Info](#)

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rds-ca-rsa2048-g1 (default)

Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

5432

Creating a database - selecting options

CRITICAL - Make sure 'Password authentication' is selected

Uncheck the 'Turn on performance Insights' check box

Expand 'Additional configuration'

Database authentication

Database authentication options [Info](#)

☒ Password authentication
Authenticates using database passwords.

☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.

☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Monitoring

☒ Turn on Performance Insights

Retention period for Performance Insights [Info](#)
7 days (free tier)

AWS KMS key [Info](#)
(default) aws/rds

Account
730335205353

KMS key ID
ba0e88f7-ddb2-4f2c-99b5-af5707e53d35

 You can't change the KMS key after enabling Performance Insights.

☐ Turn on DevOps Guru [Info](#)
DevOps Guru for RDS automatically detects performance anomalies for DB instances and provides recommendations.

▶ **Additional configuration**
Enhanced Monitoring

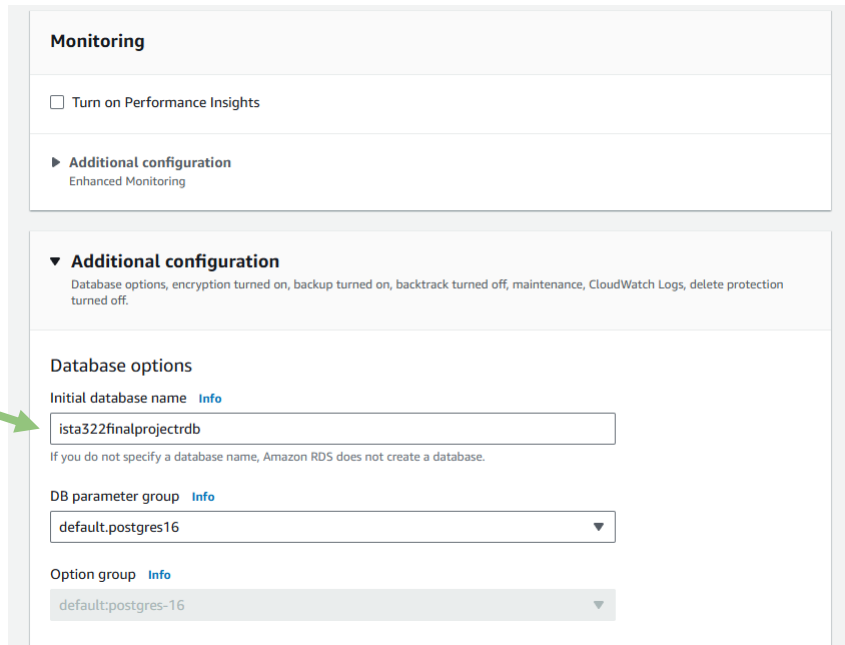
▶ **Additional configuration**
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Creating a database - selecting options

CRITICAL - Name your database. This is needed for AWS to create it and for you to access it.

*pro tip: save yourself a lot of trouble here and call this the same thing that you used as the **DB instance identifier** above. Also, avoid any special characters like hyphens or underscores.

All other options you can just leave as the default



Monitoring

☐ Turn on Performance Insights

► **Additional configuration**
Enhanced Monitoring

▼ **Additional configuration**
Database options, encryption turned on, backup turned on, backtrack turned off, maintenance, CloudWatch Logs, delete protection turned off.

Database options

Initial database name [Info](#)

If you do not specify a database name, Amazon RDS does not create a database.

DB parameter group [Info](#)

Option group [Info](#)

Creating a database - selecting options

CRITICAL – AWS will show estimated costs, but you *won't be charged unless you go over the free tier limits* described here. You'll also receive a warning when you approach the monthly limit.

Click 'Create database'

WOOOO!

Estimated Monthly costs

DB instance	13.14 USD
Storage	2.30 USD
Total	15.44 USD

This billing estimate is based on on-demand usage as described in [Amazon RDS Pricing](#). Estimate does not include costs for backup storage, IOs (if applicable), or data transfer.

Estimate your monthly costs for the DB Instance using the [AWS Simple Monthly Calculator](#).

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

- 750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.
- 20 GB of General Purpose Storage (SSD).
- 20 GB for automated backup storage and any user-initiated DB Snapshots.

[Learn more about AWS Free Tier.](#)

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the [Amazon RDS Pricing page](#).

 You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel

Create database

Creating a database - selecting options

It'll take a few min to create.
Good time to make sure you have
your credentials written down

You can see now I
have my new db
next to the other
ones we've been
using for the
class.

Creating database [ista322finalprojectdb](#) View credential details

Your database might take a few minutes to launch. You can use settings from [ista322finalprojectdb](#) to simplify configuration of [suggested database add-ons](#) while we finish creating your DB for you.

RDS > Databases

Databases (4) Group resources Modify Actions Restore from S3 Create database

Filter by databases

DB identifier	Status	Role	Engine	Region & AZ	Size	Recommendations	CPU	Current activity	Maintenance	VPC
ista322finalprojectdb	Creating	Instance	PostgreSQL	-	db.t3.micro		-		none	vpc-09e48b97a597b7b97
moviesdb	Available	Instance	PostgreSQL	us-east-1f	db.t3.micro	3 Informational	3.30%	0 Connections	none	vpc-09e48b97a597b7b97
nb6coviddata	Available	Instance	PostgreSQL	us-east-1a	db.t3.micro		3.43%	0 Connections	none	vpc-09e48b97a597b7b97
ticketsdb	Available	Instance	PostgreSQL	us-east-1c	db.t3.micro	3 Informational	3.57%	2 Connections	none	vpc-09e48b97a597b7b97

Need to change our security credentials

- Our database is up and running. Now we just need to change our security credentials to let any ip address with the password access it.
- **If you don't change this I won't be able to grade your homework... that means a zero.**
- **Seriously, you must do this step**

Modifying security credentials

If you click on the database from the console it'll take you to the console for the individual database

Databases (4)

Filter by databases

DB identifier	Status	Role	Engine	Region & AZ	Size	Recomm
ista322finalprojectrdb	Creating	Instance	PostgreSQL	-	db.t3.micro	
moviesdb	Available	Instance	PostgreSQL	us-east-1f	db.t3.micro	3 Info
nb6coviddata	Available	Instance	PostgreSQL	us-east-1a	db.t3.micro	
ticketsdb	Available	Instance	PostgreSQL	us-east-1c	db.t3.micro	3 Info

To modify your security group click the link under 'VPC security groups'

RDS > Databases > ista322finalprojectrdb

ista322finalprojectrdb

Summary

DB identifier	Status	Role	Engine
ista322finalprojectrdb	Creating	Instance	PostgreSQL
CPU	Class	Current activity	Region & AZ
-	db.t3.micro		us-east-1f

Connectivity & security

Endpoint	Availability zone	VPC security groups
-	us-east-1f	default (sg-0e0c2b05cd58a5c2d)
Port	VPC	Active
-	vpc-09e48b97a597b7b97	Publicly accessible
	Subnet group	Yes
	default-vpc-09e48b97a597b7b97	Certificate authority Info
	Subnets	rds-ca-rsa2048-g1

Click 'Create security group'

Security Groups (1/2) [Info](#)

Q Filter security groups

↻

Actions ▾

Create security group

☰

Name ▾

Security group ID ▾

Security group name ▾

VPC ID ▾

Description ▾

Owner ▾

Inbound rules count ▾

Outbound rules co... ▾

☐

-

sg-0bfb3b71a43f672aa

Test

[vpc-4e72d828](#)

Allows students

044959304225

4 Permission entries

1 Permission entry

☒

-

sg-3b202d71

default

[vpc-4e72d828](#)

default VPC security gr...

044959304225

1 Permission entry

1 Permission entry

Choose a name and description

First rule will already be there. Change it to 'All TCP' and 'Anywhere-IPv4'

Click 'Add rule'

Change second rule to 'All TCP' and 'Anywhere-IPv6'

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)
Myowngroup
Name cannot be edited after creation.

Description [Info](#)
blah blah

VPC [Info](#)
vpc-4e72d828

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info
All TCP	TCP	0 - 65535	Anywhere-IPv6	
All TCP	TCP	0 - 65535	Anywhere-IPv4	

[Add rule](#)

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info
All traffic	All	All	Custom	

[Add rule](#)

Tags - optional [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Create security group](#)

Do the same for
the outbound
rules

⚠ Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ✕

Outbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All TCP ▼	TCP	0 - 65535	Anywhere-IPv4		Delete
All TCP ▼	TCP	0 - 65535	Anywhere-IPv6		Delete

⚠ Rules with destination of 0.0.0.0/0 or ::/0 allow all IP addresses to leave the instance. We recommend setting security group rules to leave the instance from known IP addresses only. ✕

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

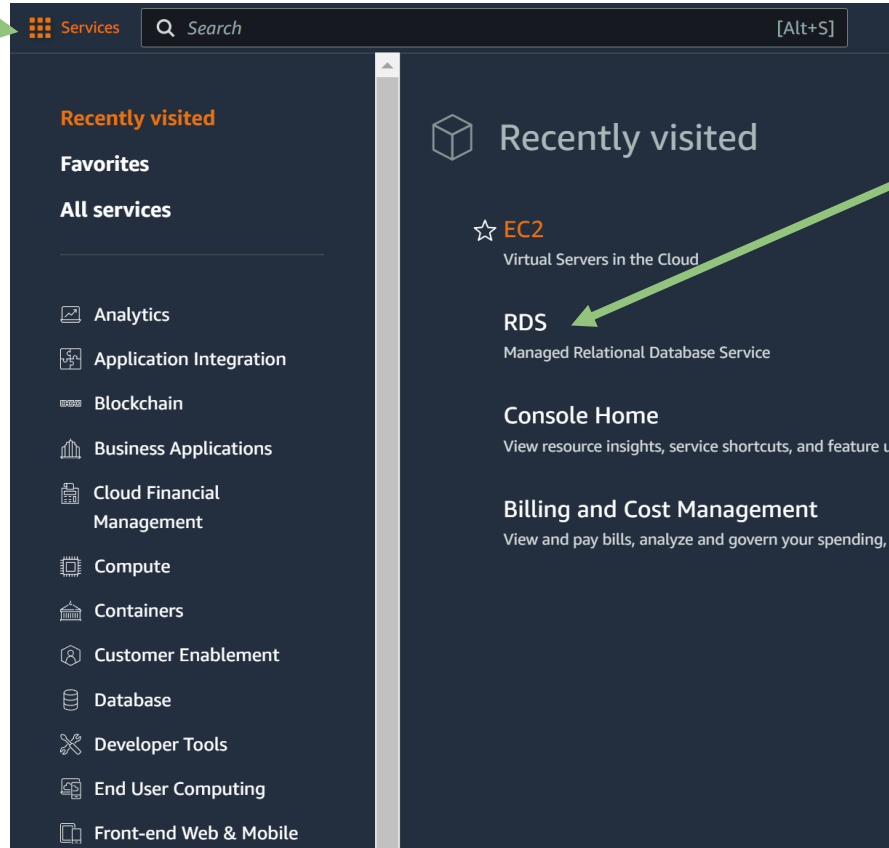
You can add up to 50 more tags.

Cancel

Finally, click
'Create security
group'

Going back to our database

Click 'services'



Then click 'RDS' which should now be in Recently visited to go back to our database dashboard.

Going back to our database

Again, we can see
I have 4 DB
instances
running. Click
on 'DB Instances

Resources

You are using the following Amazon RDS resources in the US East (N. Virginia) region (used/quota)

DB Instances (4/40)

Allocated storage (0.08 TB/100 TB)

[Increase DB instances limit](#)

DB Clusters (0/40)

Reserved instances (0/40)

Snapshots (7)

Manual

DB Cluster (0/100)

DB Instance (0/100)

Automated

DB Cluster (0)

DB Instance (7)

Recent events (18)

Event subscriptions (0/20)

Parameter groups (1)

Default (1)

Custom (0/100)

Option groups (1)

Default (1)

Custom (0/20)

Subnet groups (1/50)

[Supported platforms](#) VPC

Default network vpc-09e48b97a597b7b97

Going back to our database

Once there click on our hw database again.

Databases (4)

Filter by databases

	DB identifier ▲	Status ▼	Role ▼	Engine ▼	Region & AZ ▼	Size ▼	Recommendations
☐	ista322finalprojectrdb	✓ Available	Instance	PostgreSQL	us-east-1f	db.t3.micro	
☐	moviesdb	✓ Available	Instance	PostgreSQL	us-east-1f	db.t3.micro	3 Informational
☐	nb6coviddata	✓ Available	Instance	PostgreSQL	us-east-1a	db.t3.micro	
☐	ticketsdb	✓ Available	Instance	PostgreSQL	us-east-1c	db.t3.micro	3 Informational

Click 'Modify'

ista322finalprojectrdb



Modify

Actions ▼

Summary

DB identifier
ista322finalprojectrdb

CPU
 4.99%

Status
✔ Available

Class
db.t3.micro

Role
Instance

Current activity
 0 Connections

Engine
PostgreSQL

Region & AZ
us-east-1f

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Recommendations

Connectivity & security

Scroll down to
'Connectivity'

Click the 'X' to
delete the
default security
group

Click the pull-
down menu and
select the
security group
you just created

If everything is
right, you should
now only have a
box with your
security group
name here

The image shows three sequential screenshots of the AWS IAM console's 'Connectivity' section, illustrating the process of deleting the default security group and selecting a custom one.

Top Screenshot: The 'Connectivity' section is shown with the 'Network type' set to 'IPv4'. The 'DB subnet group' is 'default-vpc-09e48b97a597b7b97'. The 'Security group' dropdown is set to 'default'. A green arrow points from the 'Scroll down to 'Connectivity'' instruction to this section.

Middle Screenshot: The 'Security group' dropdown menu is open, showing a list of security groups: 'default', 'Myowngroup', 'Test', and 'default'. A green arrow points from the 'Click the pull-down menu and select the security group you just created' instruction to the 'Myowngroup' option.

Bottom Screenshot: The 'Security group' dropdown is now closed, and 'Myowngroup' is selected. A green arrow points from the 'If everything is right, you should now only have a box with your security group name here' instruction to the 'Myowngroup' box.

Scroll down to the bottom and click 'Continue'

044959304225
KMS key ID
12fba478-9013-459c-8dba-2fd390b1cb34

 You can't change the KMS key after enabling Performance Insights.

Maintenance
Auto minor version upgrade [Info](#)

☒ **Enable auto minor version upgrade**
Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

DB instance maintenance window
The weekly time range during which system maintenance can occur.

Start day Monday **Start time** 12 : 00 UTC **Duration** 0.5 hours

Deletion protection

☐ **Enable deletion protection**
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Cancel Continue

Choose 'Apply immediately'

Click 'Modify DB instance'

Modify DB instance: ista322moviesrdb

Summary of modifications

You are about to submit the following modifications. Only values that will change are displayed. Carefully verify your changes and click Modify DB instance.

Attribute	Current value	New value
Security group	default	Myowngroup

Scheduling of modifications

When to apply modifications

☐ Apply during the next scheduled maintenance window
Current maintenance window: September 27, 2021 05:00 - 05:30 UTC-7

☒ **Apply immediately**
The modifications in this request and any pending modifications will be asynchronously applied as soon as possible, regardless of the maintenance window setting for this database instance.

Cancel

Back

Modify DB instance

Going back to our database

Now you need to copy your endpoint.

This is the last part we need to connect. We already have the database name, password, username, and port!

Wait until the status updates to 'Available' (you might need to refresh your browser)

[RDS](#) > [Databases](#) > ista322finalprojectrdb

ista322finalprojectrdb

Summary

DB identifier
ista322finalprojectrdb

CPU
 4.15%

Status
 Available

Class
db.t3.micro

Role
Instance

Current activity
 0 Connections

Engine
PostgreSQL

Region & AZ
us-east-1f

Connectivity & security

Monitoring

Logs & events

Configuration

Maintenance & backups

Tags

Recommendations

Connectivity & security

Endpoint & port

Endpoint

 ista322finalprojectrdb.chwicm24ose3.us-east-1.rds.amazonaws.com

Port
5432

Networking

Availability Zone
us-east-1f

VPC
[vpc-09e48b97a597b7b97](#)

Subnet group
[default-vpc-09e48b97a597b7b97](#)

Subnets

Security

VPC security groups
[default \(sg-0e0c2b05cd58a5c2d\)](#)
 Active

Publicly accessible
Yes

Certificate authority [Info](#)
[rds-ca-rsa2048-g1](#)

Connecting to our database

Let's connect!

- I updated my `get_conn_cur` function with the credentials for the new database.
- Yes, I really did make my password 'password'



The screenshot shows a Jupyter Notebook window titled 'db_setup.ipynb'. The interface includes a top menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu is a toolbar with '+ Code' and '+ Text' buttons. The notebook content area displays a Python function `get_conn_cur()` that uses `psycopg2` to connect to an AWS RDS database. The function takes no arguments and returns a connection object and a cursor. Below the function definition, the function is called, and the output is shown as a connection object.

```
[5] import psycopg2
```

```
def get_conn_cur(): # define function name and arguments (there aren't any)
    # Make a connection
    conn = psycopg2.connect(
        host="test-hw-db.cnt7cm8tgir1.us-west-1.rds.amazonaws.com",
        database="hw3_db",
        user="postgres",
        password="password",
        port='5432')

    cur = conn.cursor() # Make a cursor after

    return(conn, cur) # Return both the connection and the cursor
```

```
conn, cur = get_conn_cur()
```

```
[4] conn
```

```
<connection object at 0x7f46b3703508; dsn: 'user=postgres password=xxx dbname=hw3_db host=test-hw-db.cnt7cm8tgir1.us-west-1.rds.amazonaws.com port=5432', closed: 0>
```

We run the function and store our connection and cursor

We can call `conn` which then gives us a response that we're connected to our new database!

Wooooo, you now have a database on AWS!